JESSICA WAN

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EDUCATION:

Brown University, *Providence, RI | Master of Science, Visual Computing* (Fall 2024 - May 2025)
Brown University, *Providence, RI | Bachelor of Science, Computer Science* (Fall 2020 - May 2024)

WORK EXPERIENCE:

- **C++ Software Engineer,** *Dassault Systemes – Johnston, RI* (June 2025 - Present)
  - Ship critical fixes and core optimizations to enterprise clients, reducing application latency and improving reliability.
  - Packaged hypervisor codebase into portable modules to facilitate efficient resource management.
  - Lead development of JavaScript interaction tests simulating end users, increasing test coverage by 50%.
  - Develop Python trace analysis tool to identify performance gaps, automating a previously time-intensive task for the team.
  - Lead design of internal LLM to assist with error triage and expedite risk assessment.
- **Fullstack Software Engineer Intern,** *LI-COR (Formerly Onset Data Loggers) – Bourne, MA* (May 2024 - August 2024)
  - Engineered MQTT handler processing 10k+ IoT messages/day, accelerating device-to-cloud communication.
  - Built full-stack cloud platform to consolidate and manage firmware files, significantly reducing manual configuration time.
  - Prototyped a weather map dashboard with Leaflet and GIS APIs, enabling real-time user interaction with live weather data.
- **Head Teaching Assistant,** *Brown University CS – Providence, RI* (October 2023 - May 2024)
  - Led 14-person TA team for 100+ students, managing grading, office hours, and course materials for an upper-level CS course.
  - Redesigned 6 core projects to more accurately reflect modern industry UI/UX principles.
  - Coordinated portfolio reviews with 11 industry professionals, to provide practical feedback for students.
- **Frontend Software Engineer Intern,** *LI-COR (Formerly Onset Data Loggers) – Bourne, MA* (May 2023 - August 2023)
  - Created a notification interface for users to monitor various weather sensors in line with the company's new design library.
  - Overhauled product sorting functionality as part of company-wide user interface redesign.
  - Improved middleware API efficiency as a part of the core engineering team, reducing overall latency for end users.
- **Software Consultant**, *Brown School of Public Health Survey Research Center – Providence, RI* (May 2022 - August 2022)
  - Built a JS/Qualtrics API solution for district-wide high school surveys, deployed across multiple Rhode Island school districts.
  - Engineered security features such as idle detection, screen lock, and access code authentication to protect sensitive student data.
- **Founder & Content Creator,** *Independent Art Business – tiktok.com/@minghuaa* (January 2020 - December 2023)
  - Grew a digital art brand to 170k+ followers, gaining millions of views and engagement through A/B testing and other analytics.
  - Sold 300+ digital prints and custom merchandise generating $1.6k profit in 3 months.
  - Secured official brand partnerships with several digital drawing tablet companies such as Huion and XPPen.

RECENT PROJECTS:

- **Syntax App**
  - Shipped a full-stack React speed-typing app for practicing code language syntax, leading UX design and frontend development.
  - Sourced sample code snippets from open-source repositories, integrating GPT-3 API to explain the significance of the code.
  - Engaged user incentive through personal record tracking and a community leaderboard.
- **Brown UI/UX Course Website**
  - Designed and regularly updated the CS1300 (Introduction to UI/UX) course website, serving 100+ students per semester.
- **Kiwi: Product Design/UX Engineering (YC Social Music Platform)**
  - Prototyped a mobile minimum viable product with 8 core user flows, creating a modular visual component system.
  - Ran 20+ user tests and interviews to improve app usability and user experience.
  - Delivered a developer-ready Figma handoff with accessibility annotations, edge case documentation, and a full style guide.
- **Origin Two (2D Game Engine)**
  - Built a Java game engine with a reusable UI and sound toolkit, animation cycling, and physics and collision systems.
  - Shipped 3 games with additional procedural level loading and NPC AI.
- **Ninja Mouse (3D Game Engine)**
  - Developed a C++ game engine with first and third person control, collision detection, custom shaders, and AI pathfinding.
  - Designed a component-based system to allow game-specific customizations.

SKILLS:

- **Computer Languages:** React.js, Vue.js, Python, Node.js, HTML/CSS, Typescript/JS, TensorFlow, Scipy, C++, Java, Shell
- **Tools:** Figma, Adobe, Git, Docker, Kubernetes, Unreal, Maya, Blender
- **Skills:** CI/CD, Scrum, Kanban, Project Management, Atlassian/Jira, Agile Methodologies